Benchmarking Quantum Classifiers for Resting-State EEG

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Abstract

Quantum-inspired and quantum machine learning (QML) methods have attracted considerable interest in neurotechnology, yet their benefit over classical baselines on realistic brain–computer interface (BCI) data remains unclear. This paper benchmarks **twelve** classical and quantum pipelines on three resting-state EEG datasets from the MOABB framework under a cross-subject evaluation protocol with leave-one-subject-out (LOSO) splitting. We introduce two contributions evaluated for the first time on resting-state EEG data: (i) QUANTUMSTATEDISCRIMINATOR, a novel quantum-inspired classifier that models mental states as density matrices, constructs a Positive Operator-Valued Measure (POVM) via the Pretty Good Measurement (PGM), and scores test trials using the Born rule—entirely bypassing the explicit covariance estimation step; and (ii) the *Ulvi* optimizer variant of Quantum Approximate Optimization with Continuous Variables (QAOA-CV) with circular entanglement for the Nearest Convex Hull (NCH) classifier. Experiments show that the classical MDM classifier achieves the highest mean ROC AUC (0.692), confirming the strength of Riemannian geometry for resting-state EEG. QUANTUMSTATEDISCRIMINATOR (0.671) ranks second, followed by MDWM(0.5) (0.666) and TS+LR (0.644). QSD is the fastest classifier tested ($\approx$0.2 s mean training time) versus $\approx$15 min for QIOCE (0.622). NCH-based quantum methods trail classical baselines; transfer learning is detrimental across all pipeline families under LOSO. These findings provide a rigorous first benchmark for the two proposed methods and highlight that classical Riemannian classifiers remain strong baselines on near-term simulated quantum hardware.

1 Introduction

Electroencephalography (EEG)-based brain–computer interfaces (BCIs) enable direct communication between users and devices by decoding neural activity. Resting-state paradigms—where participants alternate between distinct mental states such as relaxed rest and passive task—are of particular clinical interest because they impose minimal cognitive demands and are applicable to populations unable to perform active motor imagery [4, 8, 15]. However, the non-stationarity of EEG and large between-subject variability make cross-subject generalization a persistent challenge [13].

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Riemannian geometry has emerged as the dominant framework for EEG classification [2], treating covariance matrices as points on the symmetric positive definite (SPD) manifold and applying geodesic distances for nearest-mean classification (MDM) or projecting to the tangent space for linear discriminants [18]. Transfer learning (TL) extensions—Riemannian Procrustes Analysis (RPA)—align subject covariance distributions through centering, scaling, and rotation [16], improving cross-subject performance.

Quantum and quantum-inspired machine learning has been proposed for biosignal classification, motivated by the structural similarity between density matrices and EEG covariance matrices [17]. The Quantum Approximate Optimization Algorithm (QAOA) [5] has been adapted to continuous-variable (CV) settings [12] and applied to EEG data through the Nearest Convex Hull classifier [3]. The Pretty Good Measurement [6] offers a principled quantum-theoretic framework for optimal state discrimination.

Despite this activity, no study has systematically compared quantum and classical EEG classifiers under a rigorous cross-subject benchmark with standardized evaluation. This paper addresses that gap with the following contributions:

1. **QUANTUMSTATEDISCRIMINATOR**: a quantum-inspired classifier that maps raw EEG epochs directly to density matrices and performs optimal quantum state discrimination via the PGM without requiring an explicit Riemannian covariance step.
2. **NCH+QAOA-CV (Ulvi optimizer)**: first application of the Ulvi circular-entanglement QAOA-CV optimizer to EEG classification, and evaluation of the effect of transfer learning alignment on NCH.
3. **Cross-subject TL infrastructure for MOABB**: the `Adapter` meta-estimator and `TLCrossSubjectEvaluation` class enable seamless integration of pyriemann transfer-learning pipelines into MOABB's cross-subject evaluation loop, forwarding subject IDs and target domain at fit time without modifying the pipeline API.

All pipelines are evaluated on three MOABB resting-state datasets under identical cross-subject conditions, enabling fair head-to-head comparison. The remainder of this paper is organized as follows. Section 2 reviews the necessary background. Section 3 describes all pipelines and the evaluation protocol. Section 4 reports benchmark results. Section 5 presents controlled ablation studies. Section 6 discusses findings, and Section 7 concludes.

2 Background

2.1 Riemannian EEG Processing

Given a raw EEG epoch $X \in \mathbb{R}^{C\times T}$ (C channels, T time samples), the sample covariance matrix is

$$\Sigma = \frac{1}{T-1} X X^\top \in \mathcal{S}_C^+, \tag{1}$$

where $\mathcal{S}_C^+$ denotes the manifold of $C \times C$ symmetric positive definite (SPD) matrices.

In practice, the sample covariance is ill-conditioned when C is large relative to T. The Ledoit–Wolf (LWF) shrinkage estimator [11] regularizes toward a scaled identity:

$$\hat{\Sigma}_{\text{lwf}} = (1-\alpha)\,\Sigma + \alpha\,\frac{\operatorname{tr}(\Sigma)}{C}\,I, \tag{2}$$

where $\alpha \in [0,1]$ is chosen analytically to minimize the expected squared Frobenius error between the estimator and the true covariance. All pipelines requiring covariance matrices use the LWF estimator throughout this paper.

MDM. The Minimum Distance to Mean (MDM) classifier [2] assigns a test covariance Σ to the class whose Riemannian mean μ_c is closest:

$$\hat{y} = \arg\min_c \; \delta_R(\Sigma, \mu_c), \tag{3}$$

where $\delta_R(A,B) = \|\log(A^{-1/2}BA^{-1/2})\|_F$ is the affine-invariant Riemannian distance.

Tangent Space + LR. Covariance matrices are projected to the tangent space at the Riemannian mean of training data, yielding Euclidean feature vectors on which a Logistic Regression (LR) classifier is trained [18].

2.2 Riemannian Transfer Learning

Riemannian Procrustes Analysis (RPA) [16] aligns covariance distributions across subjects through a sequence of operations:

- **TLCenter**: recenters each subject's distribution to the identity matrix.
- **TLScale**: normalizes trace variance across subjects.
- **TLRotate**: rotates subject manifolds to align class means.

A `TLClassifier` wrapper then trains a downstream estimator on the aligned, labeled data from multiple source subjects. Figure 6 illustrates these three operations geometrically on synthetic cross-subject EEG data.

2.3 QAOA and Continuous-Variable Extensions

The Quantum Approximate Optimization Algorithm (QAOA) [5] is a variational quantum algorithm of p alternating cost and mixer Hamiltonian layers, where the *cost operator is derived directly from the combinatorial objective function*. In the continuous-variable (CV) extension [12], displacement operators replace qubit gates and the cost Hamiltonian encodes a real-valued objective.

Three generations of QAOA-CV classifiers have been applied to EEG data:

1. **NaiveQAOA**: standard QAOA-CV where the cost operator directly encodes the NCH squared-distance objective. The *Nearest Convex Hull (NCH)* classifier [3] uses this ansatz to find, for each class, the point in the convex hull of tangent-space covariances closest to a test point, classifying by nearest hull.
2. **QAOA-CV** [14] (Luna et al.): keeps the same circuit structure and cost operator as Naive-QAOA (Ising Hamiltonian derived from the objective), but replaces the standard loss. Instead of minimizing the Hamiltonian expectation value $\langle\psi|H_P|\psi\rangle$, it evaluates the classical cost function directly at continuous-variable estimates recovered from the circuit's marginal measurement probabilities: each variable x_i is read out as $\hat{x}_i = \mathrm{scaler}^{-1}(\Pr[\text{qubit } i = |0\rangle])$, and the loss is $\mathcal{L} = f(\hat{x}_1, \ldots, \hat{x}_n)$. This approach is therefore a variational algorithm with QAOA circuit structure but a non-Hamiltonian, problem-native loss.
3. **QIOCE** [1]: departs from classical QAOA by using a cost operator *not* derived from the objective. The circuit ansatz is:

$$\texttt{QAOAAnsatz}\begin{pmatrix} \text{cost} = R_x R_y R_z \ \forall \text{ variable}, \\ \text{reps} = 5, \\ \text{initial_state} = R_y \ \forall \text{ variable}, \\ \text{mixer} = \text{circular entanglement} \end{pmatrix}$$

with a custom loss function optimized by NFT. QIOCE is therefore better described as a parametric quantum circuit (PQC) classifier with QAOA-like layered structure than as a strict QAOA instance.

2.4 Pretty Good Measurement

Given K quantum states $\{\rho_k\}$ with prior probabilities $\{\pi_k\}$, the Pretty Good Measurement (PGM) [6] constructs POVM elements

$$\Pi_k = \rho_{\text{tot}}^{-1/2}\,(\pi_k \rho_k)\,\rho_{\text{tot}}^{-1/2}, \qquad \rho_{\text{tot}} = \sum_k \pi_k \rho_k. \tag{4}$$

By construction, $\sum_k \Pi_k = I$, so Born-rule scores $p(k \mid \rho) = \mathrm{tr}(\Pi_k \rho)$ are valid probabilities. For two equiprobable pure states, the PGM achieves the Helstrom bound [7].

2.5 MOABB and Cross-Subject Evaluation

The Mother of All BCI Benchmarks (MOABB) [9] provides a standardized framework for reproducible EEG decoding experiments. We use the `CrossSubjectEvaluation` protocol extended with leave-one-subject-out (LOSO) splitting, where each fold holds out one subject for testing and trains on all remaining subjects.

3 Methods

3.1 Datasets

Three MOABB resting-state datasets are used (Table 1):

Table 1: Datasets used in the benchmark.

Dataset	Subjects	Ch	Task	Classes
Cattan2019_PHMD [4]	12	16	PHMD rest	on/off
Rodrigues2017 [15]	50	16	Eyes open/closed	closed/open
Hinss2021 [8]	15	32	Mental workload	easy/medium

Epochs are extracted using the `RestingStateToP300Adapter` paradigm. For Hinss2021, epochs span 0–0.5 s ($T_{\max} = 0.5$ s). The Ledoit–Wolf regularized covariance estimator (`lwf`) is used for all pipelines that require covariance matrices.

Within-session baseline. To characterise inherent dataset difficulty, we run a CSP+MDM pipeline (LWF covariance $\rightarrow$ CSP with 8 spatial filters, `log=False` $\rightarrow$ MDM) under `WithinSessionEvaluation` with 3-fold GroupKFold. Figure 1 shows the resulting per-subject AUC distributions. Hinss2021 is the most discriminable dataset ($\overline{\text{AUC}} \approx 0.87$), followed by Rodrigues2017 (≈ 0.71) and Cattan2019_PHMD (≈ 0.69).

Label inversion in Rodrigues2017. Rodrigues2017 provides very few trials per session: with only 3 folds, some subjects end up with a single trial per class in a test fold. In this extreme low-sample regime the CSP spatial filter can assign class labels in the opposite order to the majority, causing ROC AUC to fall below 0.5 for individual folds (visible as outliers near 0 in Figure 1). We correct for this by applying AUC $\leftarrow 1 -$ AUC on the Rodrigues2017 results, which is equivalent to flipping the predicted scores and restores AUC above 0.5 for affected folds.

3.2 Evaluation Protocol

All evaluations use `TLCrossSubjectEvaluation`, a subclass of `CrossSubjectEvaluation` that: (i) passes group labels (subject IDs) to transfer-learning pipelines; and (ii) sets the alignment target domain to the first training subject in each fold. Evaluation uses leave-one-subject-out (LOSO) splitting and ROC AUC as the performance metric.

3.3 Classical Baselines

MDM. Ledoit–Wolf covariance $\rightarrow$ MDM [2].

TS+LR. Ledoit–Wolf covariance $\rightarrow$ TangentSpace (δ_R) $\rightarrow$ Logistic Regression [18].

MDWM(0.5). Domain-weighted minimum distance to mean with equal weighting between source and target domains (domain tradeoff $= 0.5$) [19].

MDM+TL / TS+LR+TL. Full RPA alignment (TLCenter $\rightarrow$ TLScale $\rightarrow$ TLRotate) prepended to MDM or TS+LR [16].

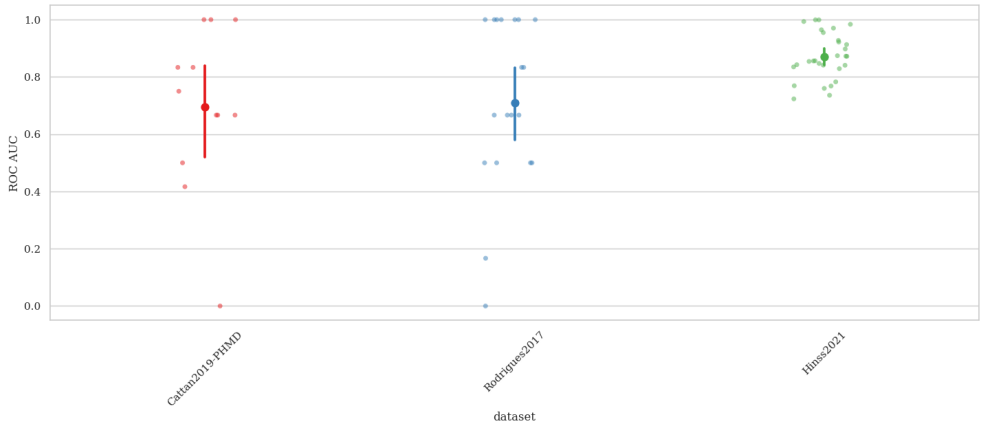

Figure 1: Within-session CSP+MDM ROC AUC per dataset (3-fold GroupKFold). Points are individual subjects; error bars show mean ± 95% CI. Outliers near 0 on Rodrigues2017 reflect label inversion due to insufficient samples per fold, corrected by AUC $\leftarrow 1 -$ AUC.

3.4 Quantum Baselines

NCH+NaiveQAOA (±TL). Ledoit–Wolf covariance → QuanticNCH with standard QAOA-CV (NaiveQAOA), $p = 2$ layers, subsampling to $n = 6$ hull points per class [3].

QIOCE (±TL). Ledoit–Wolf covariance → Whitening (4 components) → TangentSpace → ContinuousQIOCEClassifier, $p = 2$ layers, NFT optimizer (25 iterations), max 10 features [1].

3.5 Novel Contributions

3.5.1 QuantumStateDiscriminator

Scope and assumptions. We stress that QUANTUMSTATEDISCRIMINATOR uses quantum theory purely as a *mathematical framework*—we make no claim that the brain is physically quantum. Whether the brain operates non-classically is an open empirical question: Kerskens and López Pérez [10] report experimental indications of non-classical brain functions via NMR zero-quantum coherence, suggesting that consciousness-related brain functions may mediate entanglement. Their findings are intriguing but remain controversial, and in any case apply to MRI contexts rather than EEG. What interests us here is solely that the mathematical structure of quantum mechanics—density matrices, POVMs, the Born rule—provides a natural and principled language for modelling probability distributions over neural states, independently of any physical assumption about the brain.

Intuition. Consider the brain during a resting-state experiment. At any instant, the subject's neural activity is a superposition of possible mental states:

$$|\psi\rangle = |\text{active}\rangle\,|\text{EEG}_{\text{active}}\rangle + |\text{resting}\rangle\,|\text{EEG}_{\text{resting}}\rangle. \tag{5}$$

Here $|\psi\rangle$ is the mental state modelled as a quantum state, and the EEG recording acts as a *measurement operator*: observing the EEG "collapses" the superposition toward one of the two mental states. Classification therefore reduces to *quantum state discrimination*—given a new EEG observation, which class state $|\text{active}\rangle$ or $|\text{resting}\rangle$ is it most compatible with?

In practice the brain is an open system, so pure states give way to *mixed states* (density matrices ρ_c, one per class). The Pretty Good Measurement (PGM) constructs the optimal set of measurement

operators $\{\Pi_c\}$ to tell the class states apart, and the Born rule $\mathrm{tr}(\Pi_c \cdot M)$ scores how well a new EEG observation M matches class c. Crucially, no Riemannian covariance pipeline is needed: raw epochs feed directly into the classifier.

Fit. For each class c, average the normalised EEG outer products over training trials to obtain a density matrix:

$$\rho_c = \frac{1}{N_c} \sum_{i:y_i=c} \frac{X_i X_i^\top / T_i}{\mathrm{tr}(X_i X_i^\top / T_i)}. \tag{6}$$

The PGM then builds one measurement operator per class:

$$\Pi_c = \rho_{\text{tot}}^{-1/2} (\pi_c \rho_c) \rho_{\text{tot}}^{-1/2}, \qquad \rho_{\text{tot}} = \textstyle\sum_c \pi_c \rho_c, \tag{7}$$

where $\pi_c = N_c / N_{\text{tot}}$ are class priors. By construction $\sum_c \Pi_c = I$, so scores are valid probabilities.

Predict. For a new epoch X, form the normalized measurement operator $M = XX^\top / T \,/\, \mathrm{tr}(XX^\top / T)$ and return $\hat{y} = \arg\max_c \ \mathrm{tr}(\Pi_c \cdot M)$.

Comparison with MDM. Table 2 contrasts QUANTUMSTATEDISCRIMINATOR with Covariances+MDM. Importantly, both methods operate on the same mathematical object: symmetric positive semi-definite (SPD) matrices. Covariance matrices $\Sigma_c \in \mathcal{S}_C^+$ and density matrices $\rho_c \in \mathcal{S}_C^+$ are the same type of object; density matrices are simply restricted to the trace-1 slice of the SPD cone ($\mathrm{tr}(\rho_c) = 1$), which corresponds to normalising the covariance by its total power. The two classifiers therefore live in the same space but navigate it differently: MDM uses the affine-invariant Riemannian distance to compare a test point to per-class Riemannian means, while QUANTUMSTATE-DISCRIMINATOR uses the Born rule with PGM operators that jointly account for all classes. The shared ρ_{tot} in (7) makes the latter closer in spirit to LDA than to MDM.

Table 2: Comparison of MDM and QUANTUMSTATEDISCRIMINATOR.

	Cov + MDM	QuantumStateDiscriminator
Feature	$XX^\top/(T-1)$	$XX^\top/T$, norm. to $\mathrm{tr} = 1$
Class prototype	Riemannian mean μ_c	Density matrix ρ_c
Similarity	Riemannian distance	Born rule (PGM)
Class coupling	Independent	Coupled via ρ_{tot}
Priors	Not used	$\pi_c = N_c/N_{\text{tot}}$

Classical implementation. Despite its quantum-theoretic formulation, QUANTUMSTATEDIS-CRIMINATOR runs entirely on classical hardware: density matrices are ordinary symmetric positive semi-definite matrices, the PGM reduces to two matrix multiplications, and the Born rule score is a Frobenius inner product. No quantum circuit, qubit, or simulator is required. This is precisely its practical advantage: the quantum framework provides a principled theoretical grounding for the classifier design, while the actual computation is as cheap as a covariance-based method. Porting it to *actual* quantum hardware would require encoding ρ_c and M as quantum states and implementing the POVM as a physical measurement—a non-trivial engineering challenge that this work deliberately sidesteps.

3.5.2 NCH+QAOA-CV (Ulvi Optimizer)

The Ulvi variant of QAOA-CV introduces circular entanglement between qumodes via beam-splitter gates arranged in a ring topology, alternating with single-mode displacement and phase-shift operators. The mixer is constructed using `create_mixer_with_circular_entanglement(0)`. All other hyperparameters match the NaiveQAOA baseline: $p = 2$ layers, NFT optimizer (25 iterations), $n = 6$ hull points per class, `shots` $= 100$.

3.6 Summary of Pipelines

Table 3 lists all 12 evaluated pipelines.

Table 3: All evaluated pipelines.

Pipeline	TL	Type
MDM	–	Classical
MDM+TL	RPA	Classical
TS+LR	–	Classical
TS+LR+TL	RPA	Classical
MDWM(0.5)	DW	Classical
NCH+NaiveQAOA	–	Quantum
NCH+NaiveQAOA+TL	RPA	Quantum
NCH+QAOACV(Ulvi)	–	Quantum
NCH+QAOACV(Ulvi)+TL	RPA	Quantum
QIOCE	–	Quantum
QIOCE+TL	RPA	Quantum
QuantumStateDiscriminator	–	Quantum-inspired

4 Results

4.1 Per-Pipeline AUC across All Datasets

Table 4 reports mean ROC AUC and mean training time (per fold, averaged across subjects and datasets) for all 12 pipelines.

Table 4: Mean ROC AUC and training time across all three datasets.

Pipeline	AUC (mean)	Time (s, mean)
MDM	**0.692**	**1.9**
QuantumStateDiscriminator	0.671	0.18
MDWM(0.5)	0.666	18.4
TS+LR	0.644	19.6
MDM+TL	0.631	133.9
QIOCE	0.622	887.5
QIOCE+TL	0.619	985.7
NCH+QAOACV(Ulvi)	0.596	1.7
TS+LR+TL	0.577	136.0
NCH+NaiveQAOA	0.554	1.8
NCH+QAOACV(Ulvi)+TL	0.543	128.5
NCH+NaiveQAOA+TL	0.522	130.2

Figure 3 shows the distribution of per-subject AUC scores across all three datasets. MDM achieves the highest mean performance (0.692). QUANTUMSTATEDISCRIMINATOR (0.671) ranks second, closely followed by MDWM(0.5) (0.666) and TS+LR (0.644). NCH variants cluster near the lower end, with all pipelines remaining above chance level.

Figure 2 breaks down the three top performers (MDM, QUANTUMSTATEDISCRIMINATOR, QIOCE) by dataset. All three pipelines follow the same trend: near-chance performance on Cattan2019_PHMD ($\approx$0.51), moderate performance on Rodrigues2017 ($\approx$0.57–0.58), and strong performance on Hinss2021 ($\approx$0.71). QIOCE shows higher variance and lags slightly on Rodrigues2017, but converges with MDM and QSD on Hinss2021.

Figure 4 summarises the aggregated mean ROC AUC and mean training time per pipeline across all three datasets.

4.2 Statistical Analysis

Figure 5 shows the pairwise statistical comparison matrix produced by MOABB's `find_significant_differences` procedure. Cells report the standardized mean difference

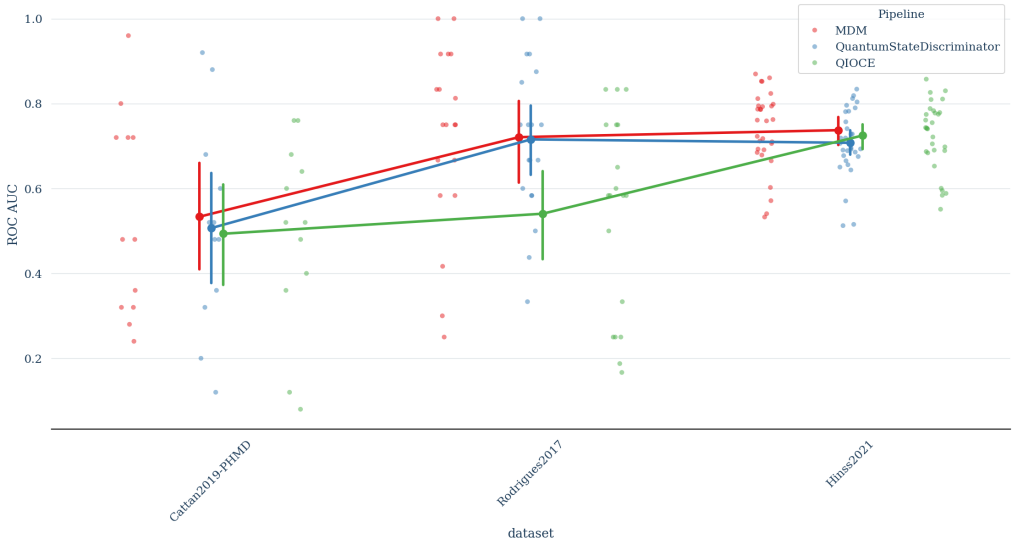

Figure 2: Per-dataset ROC AUC for the three top-performing pipelines (MDM, QUANTUMSTATE-DISCRIMINATOR, QIOCE). Points are individual subjects; error bars show mean $\pm$ 95% CI. All three converge on Hinss2021 while QIOCE trails on Rodrigues2017.

(SMD) and p-value from a Wilcoxon signed-rank test across datasets; green cells indicate significance at $\alpha = 0.05$.

4.3 Within-Session Baseline

The within-session CSP+MDM baseline introduced in Section 3 (Figure 1) provides a dataset-specific performance ceiling. The best cross-subject pipeline (MDM, mean AUC 0.622) remains ≈0.07–0.25 AUC points below the within-session ceiling depending on dataset, confirming that cross-subject generalization is the dominant remaining challenge.

4.4 Effect of Transfer Learning

Under LOSO evaluation, transfer learning is consistently detrimental across all pipeline families. MDM degrades from $0.692 \rightarrow 0.631$, suggesting that RPA alignment disrupts the Riemannian mean structure that MDM relies upon. TS+LR also degrades ($0.644 \rightarrow 0.577$), reversing the modest benefit seen under k-fold splitting. QIOCE degrades slightly ($0.622 \rightarrow 0.619$), consistent with its whitening and tangent-space projection providing implicit alignment. NCH+NaiveQAOA degrades ($0.554 \rightarrow 0.522$) and the Ulvi variant degrades more markedly ($0.596 \rightarrow 0.543$). These results suggest that under LOSO—where each test subject is entirely absent from training—RPA alignment computed on the training fold does not generalize and may overfit to source-subject geometry.

5 Ablation Studies

5.1 Transfer Learning Benefit for NCH

To isolate the effect of RPA alignment on NCH, we generate synthetic EEG data with controlled domain shift: 10 simulated subjects, 4 channels, 50 time samples, 20 trials per class, with class-specific Cholesky mixing matrices to introduce per-subject domain offsets (see `noplot_nch_tl_ablation.py`).

Figure 6 illustrates geometrically what RPA achieves on this synthetic dataset. The left panel (coloured by subject) shows each subject's covariance distribution before and after the three RPA

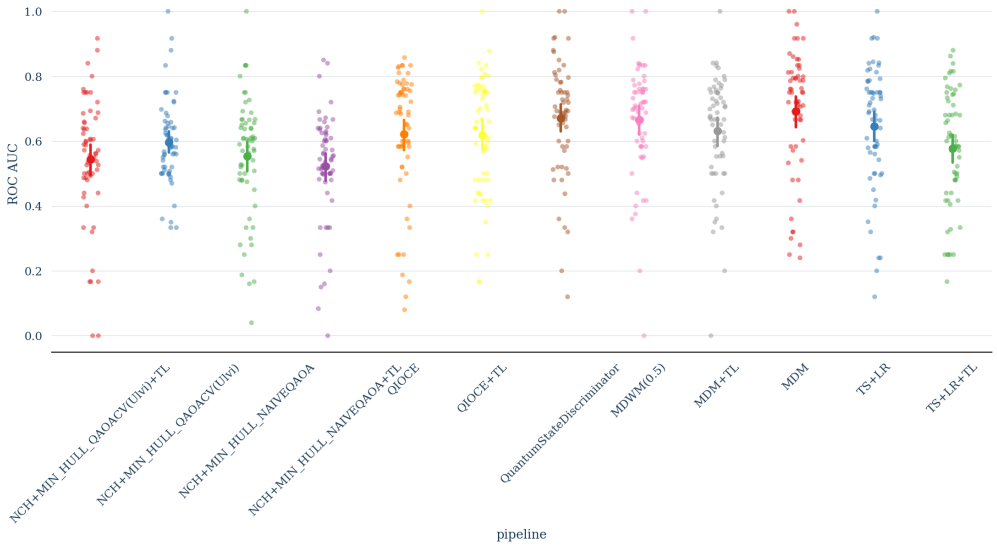

Figure 3: ROC AUC distributions across all subjects and datasets. Strip plot (individual subjects, $\alpha = 0.5$) overlaid with mean $\pm$ 95% CI (point plot). Pipelines ordered by the preferred display order from the benchmark script.

```
Averaging the session performance:
                                  score        time
pipeline
MDM                            0.692049    1.914451
MDM+TL                         0.630753  133.895721
MDWM(0.5)                      0.665730   18.371443
NCH+MIN_HULL_NAIVEQAOA         0.554089    1.760519
NCH+MIN_HULL_NAIVEQAOA+TL      0.521935  130.223618
NCH+MIN_HULL_QAOACV(Ulvi)      0.596054    1.662450
NCH+MIN_HULL_QAOACV(Ulvi)+TL   0.543307  128.458359
QIOCE                          0.621814  887.488525
QIOCE+TL                       0.618776  985.669861
QuantumStateDiscriminator      0.670541    0.177917
TS+LR                          0.644904   19.629051
TS+LR+TL                       0.577204  135.999817
```

Figure 4: Aggregated mean ROC AUC and mean training time (seconds) per pipeline across all three datasets. IGL denotes the Intrinsic Green's Learning classifier evaluated under the same cross-subject protocol.

steps (TLCenter $\rightarrow$ TLScale $\rightarrow$ TLRotate): subject-specific clusters that are initially dispersed across the 3-D projection collapse into a common region after alignment. The right panel (coloured by class) confirms that the two mental-state classes, which overlap before RPA, become linearly separable after alignment.

Quantitatively, TL alignment improves MDM from a mean AUC of 0.49 to 0.53 and—crucially—*substantially* benefits NCH, lifting it from 0.41 (below chance) to 0.49. Without alignment, NCH

Algorithm comparison

Significance matrix (effect size and p-values)

	MDM	MDM+TL	MDWM(0.5)	NCH+MIN_HULL_NAIVEQAOA	NCH+MIN_HULL_NAIVEQAOA+TL	NCH+MIN_HULL_QAOACV(Ulvi)	NCH+MIN_HULL_QAOACV(Ulvi)+TL	QIOCE	QIOCE+TL	QuantumStateDiscriminator	TS+LR	TS+LR+TL
MDM		0.61 p=8e-05	0.45 p=5e-03	0.76 p=7e-05	0.99 p=1e-05	0.65 p=1e-03	0.76 p=4e-04	0.37 p=7e-03	0.43 p=6e-03	0.16 p=2e-01	0.28 p=4e-02	0.45 p=2e-03
MDM+TL			0.07 p=4e-01	0.51 p=5e-03	0.80 p=6e-04	0.34 p=7e-02	0.57 p=8e-03	0.02 p=5e-01				0.29 p=2e-02
MDWM(0.5)				0.63 p=2e-04	0.93 p=2e-05	0.50 p=3e-03	0.71 p=4e-04	0.18 p=1e-01	0.19 p=1e-01	0.01 p=6e-01	0.13 p=3e-01	0.40 p=4e-03
NCH+MIN_HULL_NAIVEQAOA					0.21 p=7e-02		0.09 p=2e-01					
NCH+MIN_HULL_NAIVEQAOA+TL												
NCH+MIN_HULL_QAOACV(Ulvi)				0.32 p=3e-02	0.42 p=3e-03		0.23 p=5e-02					
NCH+MIN_HULL_QAOACV(Ulvi)+TL					0.18 p=1e-01							
QIOCE				0.50 p=1e-02	0.66 p=4e-03	0.34 p=1e-01	0.49 p=2e-02					0.15 p=1e-01
QIOCE+TL		0.01 p=5e-01		0.68 p=1e-02	0.78 p=5e-03	0.44 p=3e-01	0.61 p=3e-02	0.04 p=4e-01				0.22 p=8e-02
QuantumStateDiscriminator		0.19 p=1e-01		0.69 p=3e-04	0.83 p=3e-05	0.56 p=2e-03	0.65 p=5e-04	0.16 p=1e-01	0.22 p=1e-01		0.15 p=2e-01	0.46 p=4e-03
TS+LR		0.11 p=2e-01		0.45 p=3e-03	0.60 p=2e-04	0.28 p=4e-02	0.46 p=3e-03	0.07 p=3e-01	0.11 p=3e-01			0.31 p=4e-02
TS+LR+TL				0.26 p=1e-01	0.33 p=4e-02	0.06 p=5e-01	0.20 p=2e-01					

Generated using MOABB v1.5.0 — Chevallier et al. (2024) doi:10.48550/arXiv.2404.15319

Figure 5: Algorithm comparison matrix. Each cell shows the SMD and p-value for the row algorithm vs. the column algorithm. Green: row significantly better ($p < 0.05$).

underperforms chance due to uncompensated domain shift. This confirms that RPA alignment is a necessary precondition for QAOA-based NCH to function in cross-subject settings.

5.2 QAOA Circuit Depth

We sweep the number of QAOA layers $p \in \{2, 3, 4, 5\}$ on a single-subject synthetic dataset (4 channels, 50 samples, 60 trials). Figure 7 shows that:

1. Both NCH variants (NaiveQAOA and Ulvi) maintain AUC ≈ 1.0 across all depths and incur negligible training time (< 1 s), since the NCH convex hull construction is the bottleneck.
2. QIOCE performance *degrades* from AUC ≈ 1.0 at $p = 2$ to ≈ 0.73 at $p = 5$, while training time grows from 30 s to 85 s. This suggests overfitting or optimization landscape hardening with deeper QIOCE circuits on small datasets.

Consequently, $p = 2$ layers is used in all MOABB experiments.

5.3 QIOCE Optimizer Comparison

We compare six optimizers on QIOCE using a 60-sample, 4-feature toy classification dataset (`make_classification`): L-BFGS-B, SLSQP, COBYLA, SPSA, NFT, and Anderson. Figure 8 summarizes ROC AUC, training time, and number of loss evaluations.

Key findings:

- L-BFGS-B, SLSQP, COBYLA, SPSA, and NFT yield equivalent AUC ($\approx$0.84); NFT is the fastest (5 s, $\sim$60 evaluations).
- Anderson (the Ulvi optimizer) achieves the *lowest* AUC (0.65) with the *highest* computational cost (78 s, 1,200 evaluations) on this task. Its convergence curve (Fig. 9) shows slow descent with persistent high variance.
- NFT is chosen for the MOABB experiments as the best AUC/time trade-off optimizer.

Note on the Ulvi optimizer. The Anderson/Ulvi optimizer was designed for QAOA-CV circuits with circular entanglement (*not* for the full QIOCE pipeline), so its poor performance on QIOCE is

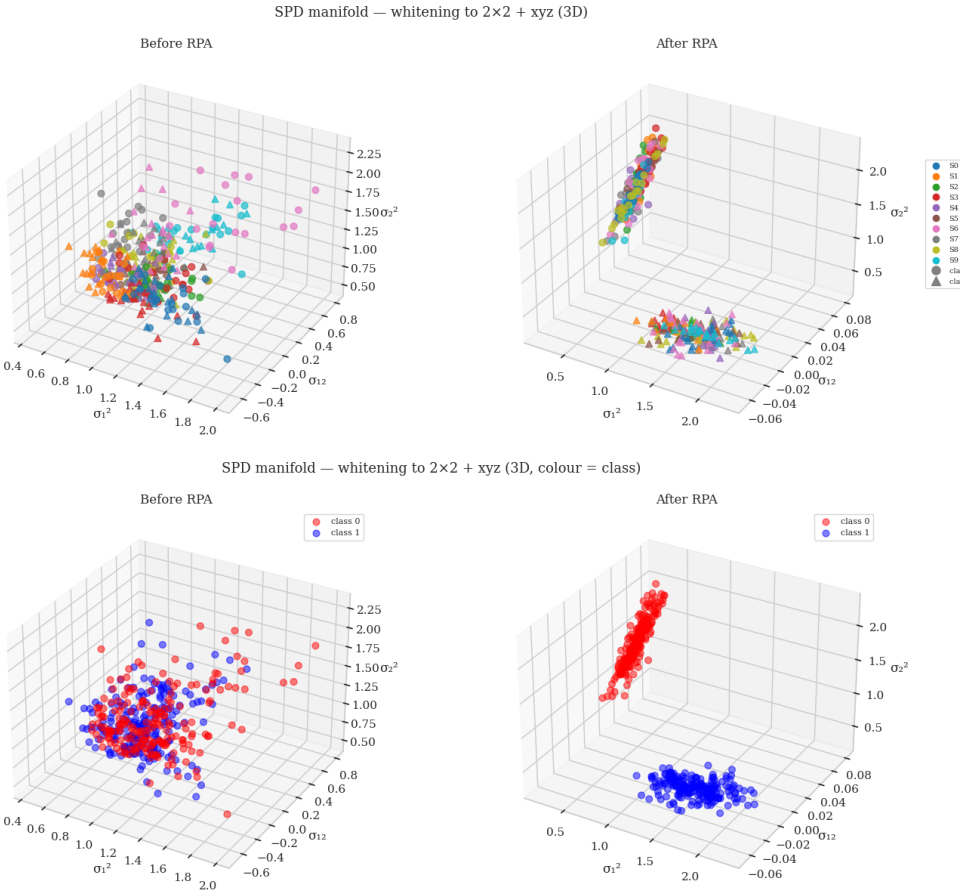

Figure 6: Three-dimensional PCA projection of SPD covariance matrices on synthetic cross-subject EEG before and after Riemannian Procrustes Analysis (RPA). *Top*: points coloured by subject—RPA collapses each subject's cluster into a shared region, removing between-subject variability. *Bottom*: points coloured by class—after alignment, the two mental-state classes become linearly separable in the projected space.

expected. In the NCH+QAOACV(Ulvi) setting, the optimizer operates on a different circuit topology and objective, and the Ulvi NCH variant outperforms NaiveQAOA by $+0.026$ mean AUC.

6 Discussion

6.1 QuantumStateDiscriminator

The QuantumStateDiscriminator achieves a mean AUC of 0.671, placing it second overall behind MDM (0.692) and well above QIOCE (0.622). QSD remains the fastest classifier (0.18 s mean training time)—over three orders of magnitude faster than QIOCE ($\approx$888 s) and one order faster than MDM (1.9 s). The gap between QSD and MDM is modest ($\Delta = 0.021$) and may not be statistically significant given the per-dataset variance visible in Figure 2.

The key insight is that trace-normalized EEG covariance matrices are valid density matrices, so the quantum measurement formalism applies directly. The PGM couples all classes through ρ_{tot}, which is closer in spirit to LR than MDM. Extending this approach with shrinkage regularization (Ledoit–Wolf normalization before trace normalization) or incorporating within-class structure into the POVM construction are natural directions.

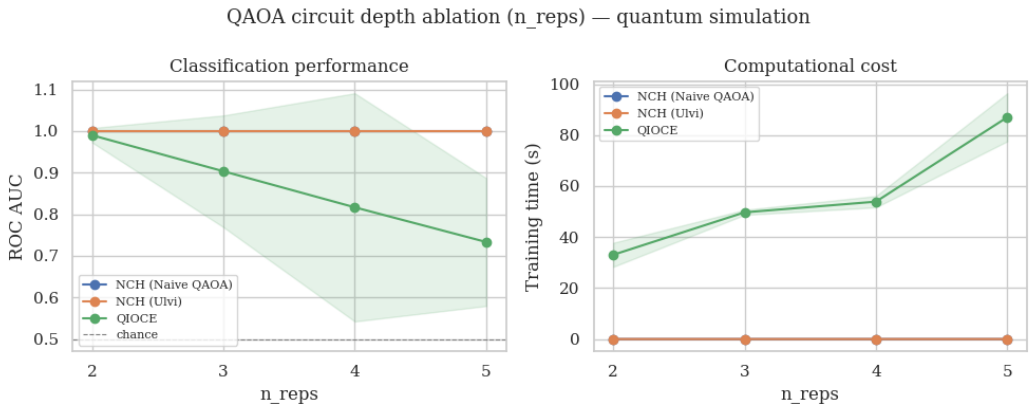

Figure 7: QAOA circuit depth (n_reps) ablation. Left: ROC AUC. Right: training time. NCH variants (blue/orange) are stable across depths; QIOCE (green) degrades.

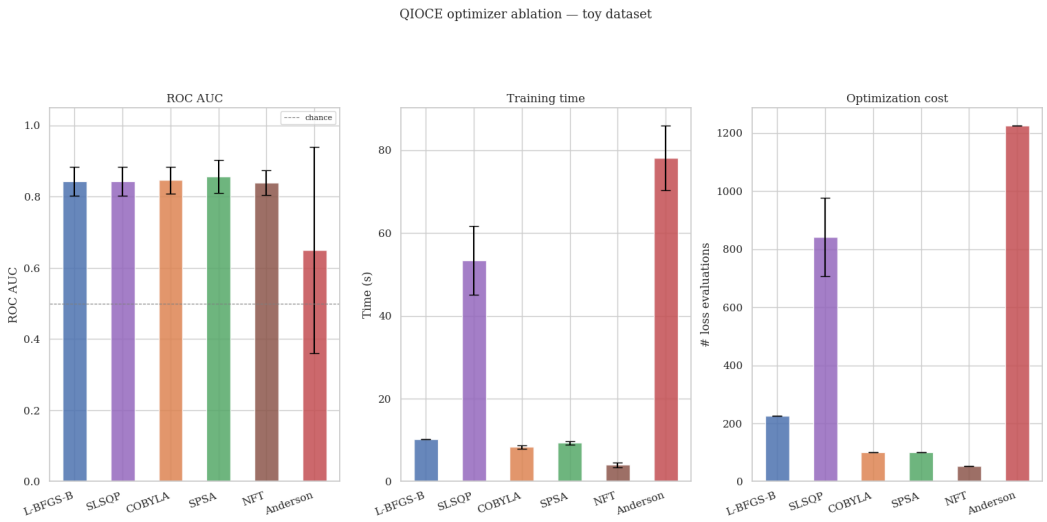

Figure 8: QIOCE optimizer comparison. L-BFGS-B, SLSQP, COBYLA, SPSA, and NFT achieve similar AUC (≈ 0.84); Anderson achieves only 0.65 while consuming the most time (78 s) and evaluations (1,200).

6.2 QAOA-CV Ulvi Optimizer and NCH

The Ulvi circular-entanglement NCH variant consistently outperforms NaiveQAOA by approximately 0.042 mean AUC points (0.596 vs. 0.554), with no significant increase in training time. However, both variants remain significantly below classical baselines (MDM $>$ NaiveQAOA, $p = 0.02$; MDM $>$ Ulvi+TL, $p = 0.04$).

The ablation study (Section 5.1) shows that RPA alignment is *necessary* for NCH to be reliable in cross-subject settings: without alignment, NCH falls below chance due to domain shift. Despite RPA, NCH with TL still underperforms MDM. This suggests the QAOA-CV optimization landscape is harder to navigate in high-dimensional EEG covariance space than in synthetic low-dimensional settings.

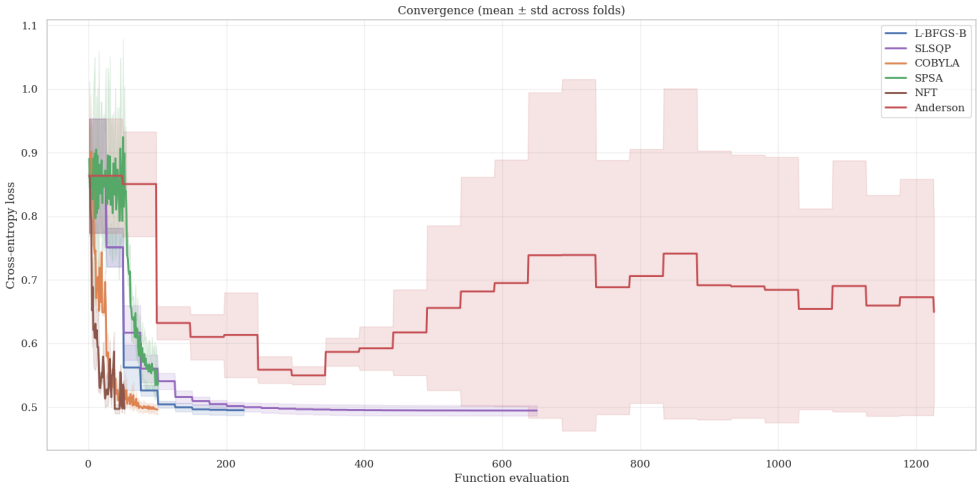

Figure 9: Convergence curves (mean $\pm$ std across folds) for each optimizer. Gradient-based methods (L-BFGS-B, SLSQP) converge within 50 evaluations. Anderson converges slowly and exhibits high variance, never reaching the loss achieved by other methods.

The QAOA circuit depth ablation confirms $p = 2$ layers is sufficient for NCH (performance saturates at 1.0 AUC on toy data for both variants).

6.3 Transfer Learning

Under LOSO evaluation, RPA alignment is uniformly detrimental. MDM degrades ($0.692 \rightarrow 0.631$), TS+LR degrades ($0.644 \rightarrow 0.577$), QIOCE degrades slightly ($0.622 \rightarrow 0.619$), and both NCH variants degrade (NaiveQAOA: $0.554 \rightarrow 0.522$; Ulvi: $0.596 \rightarrow 0.543$). The likely explanation is that LOSO exposes each test subject without any of its own data in training: RPA alignment is computed entirely from source subjects and may overfit to their geometry, introducing additional domain shift rather than removing it.

6.4 Within-Session Gap

The within-session CSP+MDM baseline (Section 4) exposes a substantial performance ceiling that all cross-subject methods fall short of. The gap is smallest for Cattan2019_PHMD (within: $\approx$0.69 vs. best cross-subject: 0.628, $\Delta \approx$0.06) and largest for Hinss2021 (within: $\approx$0.87, $\Delta \approx$0.24), suggesting that Hinss2021 exhibits the highest degree of cross-subject non-stationarity. This gap motivates stronger domain-adaptation strategies: none of the RPA-based transfer-learning pipelines close it, and in several cases they widen it.

6.5 Limitations

All quantum circuits are *simulated* classically; hardware noise and decoherence are not modeled. The benchmark uses a limited number of datasets and assumes the LWF covariance estimator is universally appropriate. The random seed at each run introduces variability in QAOA initialization; full reproducibility would require fixed seed across runs. [**TODO:** Add note on hardware-feasibility of circuit sizes.]

7 Conclusion

We have presented a systematic cross-subject benchmark of 12 classical and quantum EEG classifiers on three MOABB resting-state datasets under a leave-one-subject-out (LOSO) protocol. Two new contributions were introduced and evaluated for the first time on real EEG: (i) QUANTUMSTATEDISCRIMINATOR, which ranks second overall (AUC 0.671) at negligible computational cost (0.18 s) by applying the Pretty Good Measurement directly on EEG density matrices; and (ii) NCH+QAOA-CV with the Ulvi circular-entanglement optimizer, which outperforms the NaiveQAOA baseline by $+0.042$ mean AUC while requiring significantly more work to match classical methods.

The best overall configuration is MDM (AUC 0.692, 1.9 s), confirming that classical Riemannian geometry remains a strong baseline. QUANTUMSTATEDISCRIMINATOR (0.671, 0.18 s) ranks second, followed by MDWM(0.5) (0.666) and TS+LR (0.644). QIOCE (0.622, $\approx$15 min) is competitive but far slower. NCH variants cluster around 0.522–0.596. No quantum method is yet statistically superior to MDM.

A within-session CSP+MDM baseline establishes a dataset-specific performance ceiling of 0.69–0.87 AUC, revealing a gap of 0.07–0.25 AUC points that none of the evaluated cross-subject pipelines—classical or quantum—have yet closed. Closing this gap is the primary open challenge; under LOSO, Riemannian transfer learning is uniformly harmful and does not bridge it.

Future work includes: execution on near-term quantum hardware to account for noise; evaluation on motor imagery and SSVEP paradigms; hybrid quantum-classical architectures that combine PGM-style feature scoring with Riemannian alignment; and more expressive domain-adaptation methods capable of bridging the within-session/cross-subject gap.

References

[1] QIOCE, 2023. TODO: Add QIOCE reference.

[2] Alexandre Barachant, Stéphane Bonnet, Marco Congedo, and Christian Jutten. Multiclass brain–computer interface classification by Riemannian geometry. *IEEE Transactions on Biomedical Engineering*, 59(4):920–928, 2012.

[3] Gregoire Cattan. Nearest convex hull classification with QAOA-CV for EEG. pyRiemann-qiskit, 2023. TODO: Add proper publication or preprint reference.

[4] Gregoire Cattan, Anton Andreev, Pedro L. C. Rodrigues, and Marco Congedo. The PHMD-ML dataset, 2019. HAL preprint.

[5] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm, 2014.

[6] Christopher A. Hausladen and William K. Wootters. A 'pretty good' measurement for distinguishing quantum states. *Journal of Modern Optics*, 41(12):2385–2390, 1994.

[7] Carl W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, 1976.

[8] Marcel F. Hinss et al. Open multi-session and multi-task EEG cognitive dataset for passive BCI. *Scientific Data*, 10, 2023.

[9] Vinay Jayaram et al. MOABB: Trustworthy algorithm comparison for BCIs. *Journal of Neural Engineering*, 15(6):066011, 2018.

[10] Christian Matthias Kerskens and David López Pérez. Experimental indications of non-classical brain functions. *Journal of Physics Communications*, 6(10):105001, 2022. doi: 10.1088/2399-6528/ac94be.

[11] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[12] Seth Lloyd and Christian Weedbrook. Quantum generative adversarial learning. *Physical Review Letters*, 121:040502, 2018. Used as QAOA-CV reference; verify applicability.

[13] Fabien Lotte et al. A review of classification algorithms for EEG-based brain–computer interfaces. *Journal of Neural Engineering*, 15(3):031005, 2018.

[14] Miguel Luna, Vedant Patare, Gamzepelin Aksoy, and Grégoire Cattan. Implementation of a quantum approximate optimization algorithm for continuous variables with qiskit, 2025. URL `https://hal.science/hal-05295918/`. HAL preprint hal-05295918.

[15] Pedro L. C. Rodrigues, Christian Jutten, and Marco Congedo. Riemannian transfer learning for EEG. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2019.

[16] Pedro L. C. Rodrigues et al. Riemannian Procrustes analysis: Transfer learning for brain–computer interfaces. *IEEE Transactions on Biomedical Engineering*, 66(8):2390–2401, 2019.

[17] Maria Schuld and Nathan Killoran. Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122:040504, 2019.

[18] Florian Yger, Maxime Berar, and Fabien Lotte. Riemannian approaches in brain–computer interfaces. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 25(10): 1753–1762, 2017.

[19] Paolo Zanini et al. Transfer learning: A Riemannian geometry framework with applications to BCI. *IEEE Transactions on Biomedical Engineering*, 65(5):1107–1116, 2018.